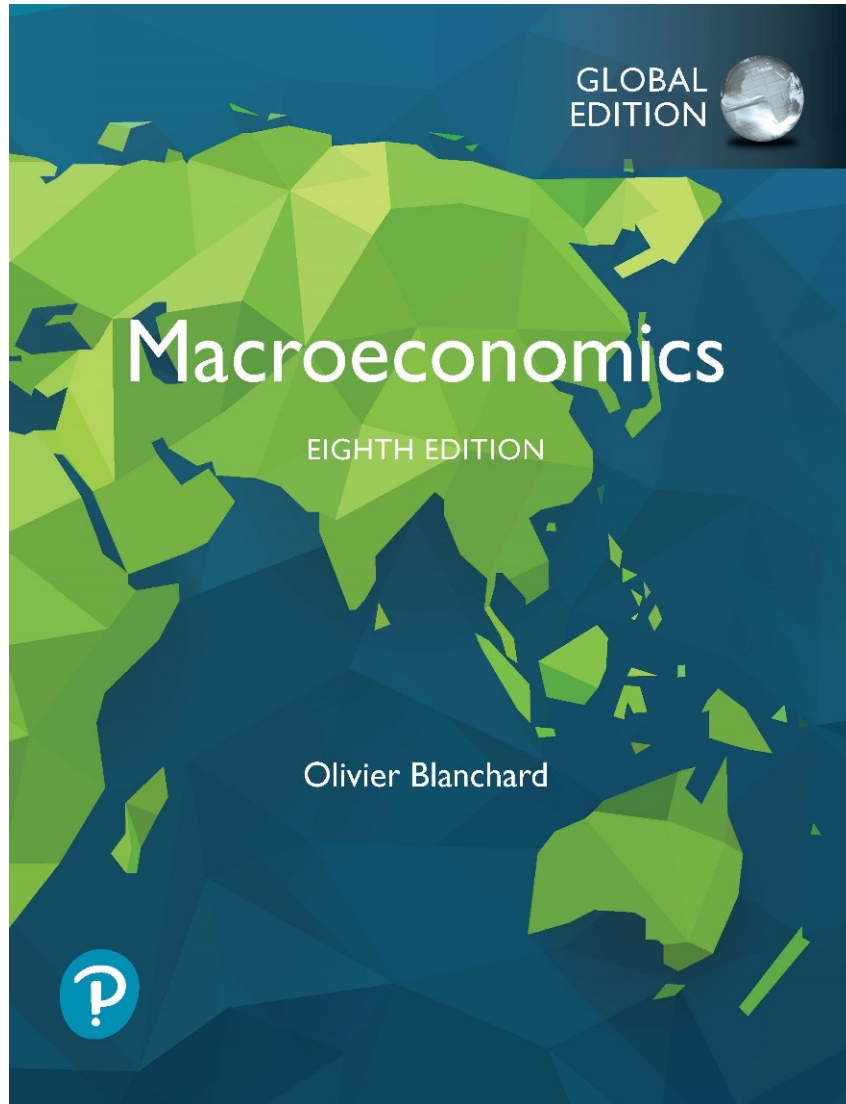


Macroeconomics

Eighth Edition, Global Edition



Chapter 8

The Phillips Curve, the Natural Rate of Unemployment, and Inflation

Chapter 8 Outline

The Phillips Curve, the Natural Rate of Unemployment, and Inflation

8.1 Inflation, Expected Inflation, and Unemployment

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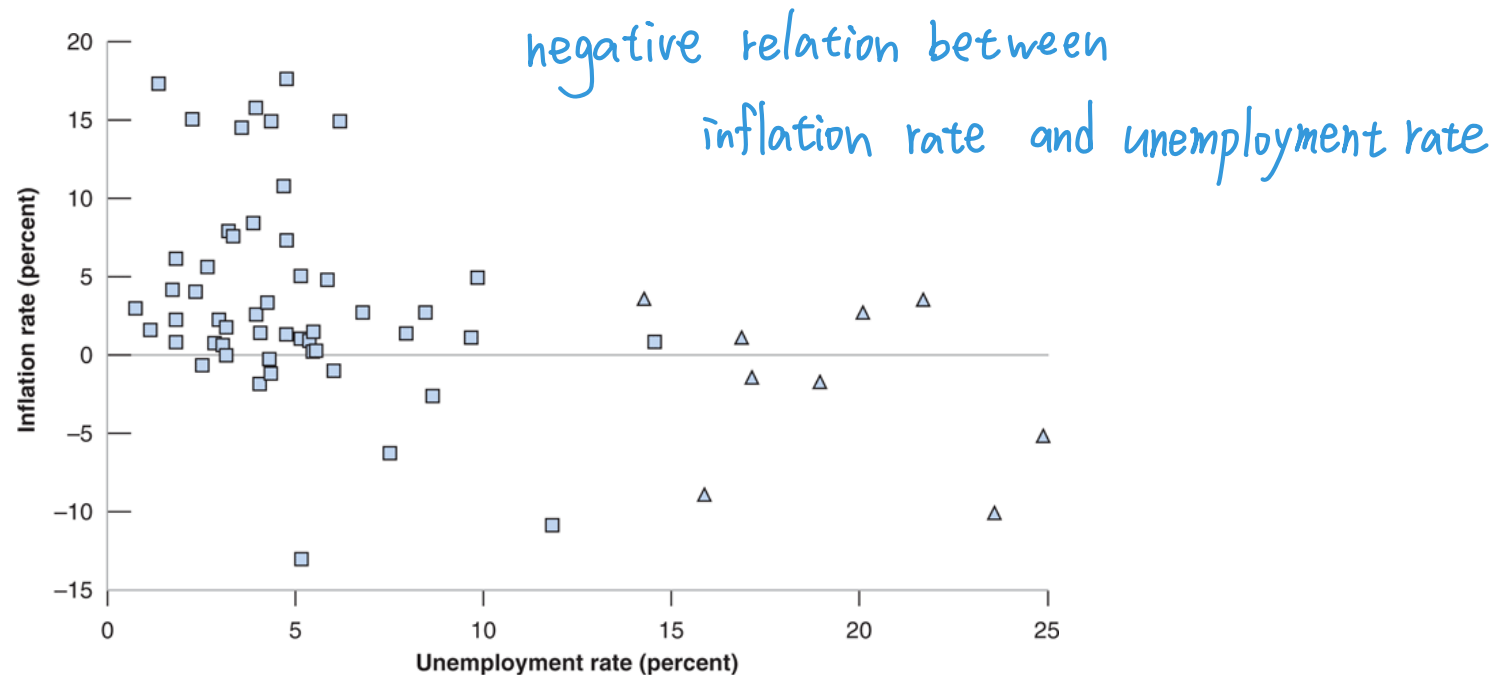
APPENDIX Derivation of the Relation Between Inflation, Expected Inflation, and Unemployment

The Phillips Curve, the Natural Rate of Unemployment, and Inflation

- In 1958, A.W. Phillips found a negative relation between inflation and unemployment.
- Two years later, Paul Samuelson and Robert Solow labeled this relation the **Phillips curve**, which became central to macroeconomic thinking and policy.

Figure 8.1 Inflation versus Unemployment in the United States, 1900–1960

During the period 1900-1960 in the United States, a low unemployment rate was typically associated with a high inflation rate, and a high unemployment rate was typically associated with a low or negative inflation rate.



Source: Based on Historical Statistics of the United States.

<http://hsus.cambridge.org/HSUSWeb/index.do>.

8.1 Inflation, Expected Inflation, and Unemployment (1 of 3)

- Recall the wage determination equation (7.1):

$$W = P^e F(u, z) \quad (7.1)$$

and the price determination equation (7.3):

$$P = (1 + m)W \quad (7.3)$$

8.1 Inflation, Expected Inflation, and Unemployment (2 of 3)

- Assume a specific form for F :

$$F(u, z) = 1 - \alpha u + z$$

so that the *relation between the price level, the expected price level, and the unemployment rate* is:

$$P = P^e(1 + m)(1 - \alpha u + z) \quad (8.1)$$

or in terms of inflation rate π and the expected inflation rate π^e :

$$\pi = \pi^e + (m + z) - \alpha u \quad (8.2)$$

8.1 Inflation, Expected Inflation, and Unemployment (3 of 3)

- *An increase in π^e leads to an increase in π .*
- *Given π^e , an increase in m , or an increase in z , leads to an increase in π .*
- *Given π^e , a decrease in u leads to an increase in π .*
- Equation (8.2) with a time index t :

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t \quad (8.3)$$

8.2 The Phillips Curve and Its Mutations (1 of 9)

- Assume $\pi_t^e = \bar{\pi}$, so that equation (8.3) becomes:

$$\pi_t = (m + z) - \alpha u_t \quad (8.4)$$

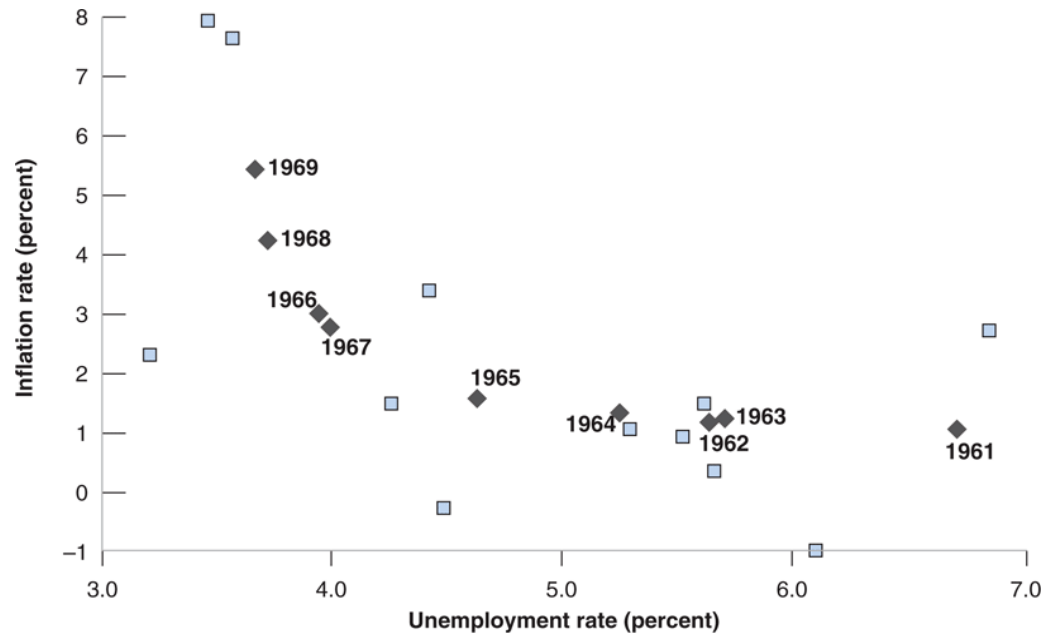
which is a negative relation between unemployment and inflation.

- This Phillips curve relation was observed in the United States in the 1960s.
- The relation vanished in the 1970s because wage setters changed the way they formed inflation expectations.

8.2 The Phillips Curve and Its Mutations (2 of 9)

Figure 8.2 Inflation versus Unemployment in the United States, 1961–1969

The steady decline in the U.S. unemployment rate throughout the 1960s was associated with a steady increase in the inflation rate.

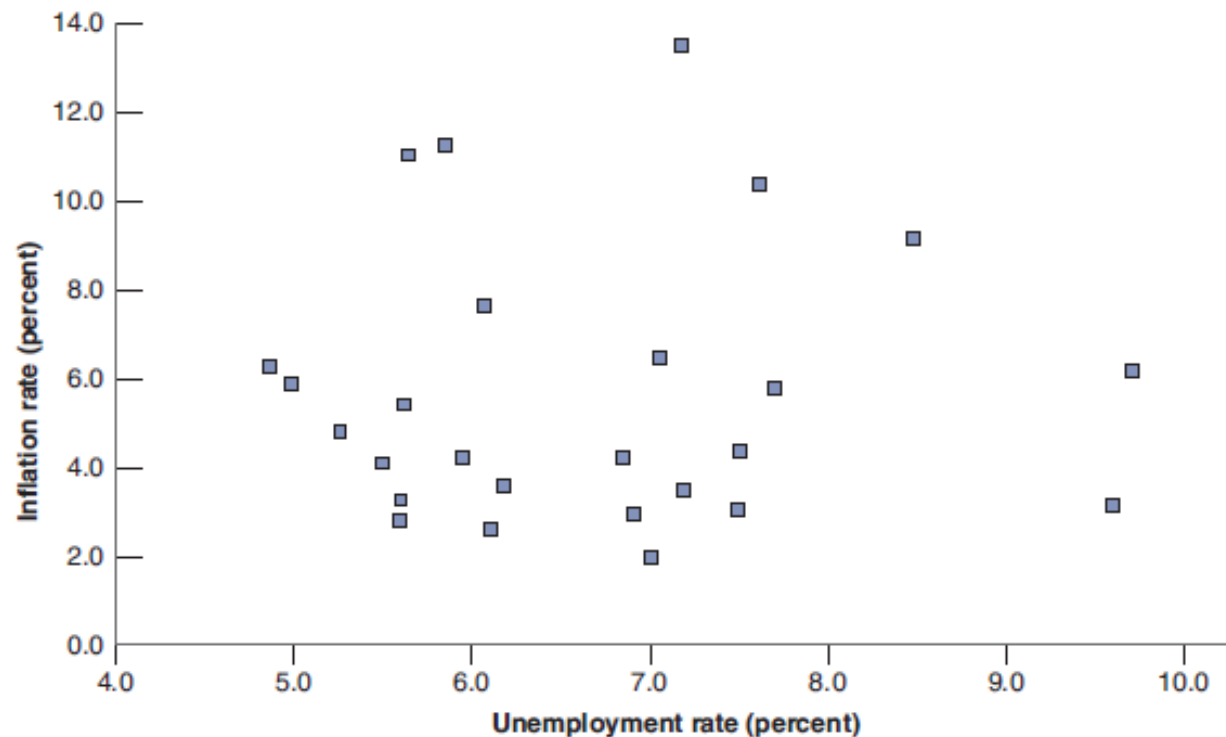


- Source: FRED: Series UNRATE, CPIAUSCL

8.2 The Phillips Curve and Its Mutations (3 of 9)

Figure 8.3 Inflation versus Unemployment in the United States, 1970–1995

Beginning in 1970 in the United States, the relation between the unemployment rate and the inflation rate disappeared.



Source: FRED: UNRATE, CPIAUSCL.

8.2 The Phillips Curve and Its Mutations (4 of 9)

- Suppose expected inflation this year depends on a constant value $\bar{\pi}$ with weight $1-\theta$, and partly on inflation last year with weight θ :

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1} \quad (8.5)$$

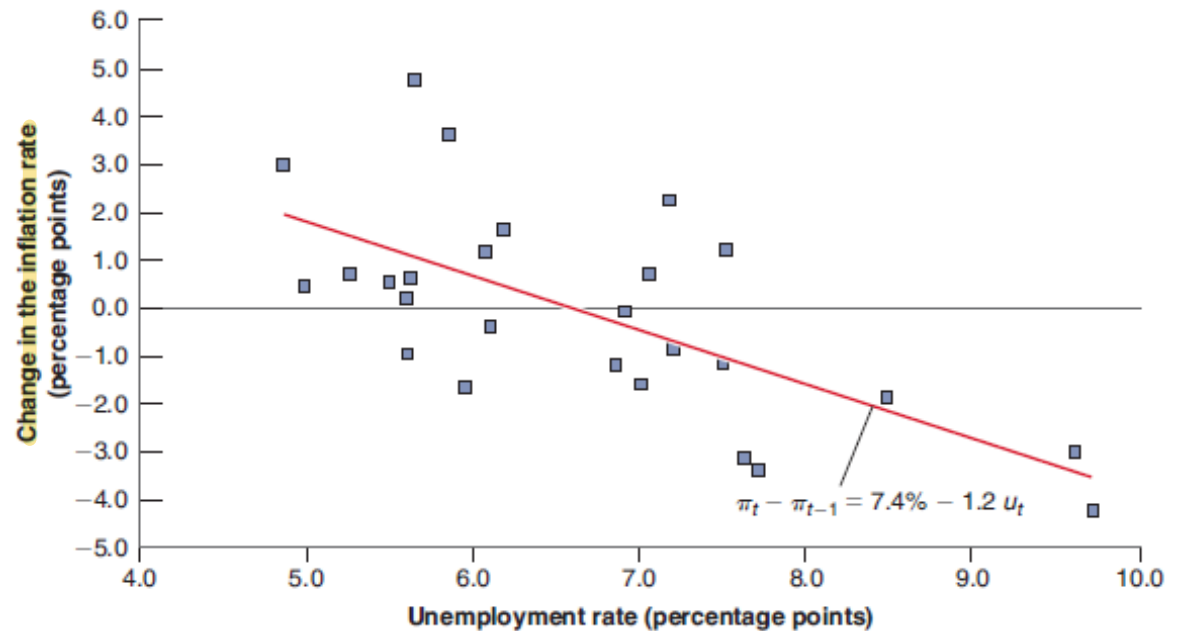
- When $\theta = 0$, $\pi_t = \bar{\pi} + (m + z) - \alpha u_t$
- When $\theta > 0$, $\pi_t = [(1 - \theta)\bar{\pi} + (m + z)] + \theta\pi_{t-1} - \alpha u_t$
- When $\theta = 1$, $\pi_t - \pi_{t-1} = (m + z) - \alpha u_t$, (8.6)

So the unemployment rate affects not the inflation rate, but rather the change in the inflation rate.

8.2 The Phillips Curve and Its Mutations (5 of 9)

Figure 8.4 Change in Inflation versus Unemployment in the United States, 1970–1995

From 1970 to 1995, there was a negative relation between the unemployment rate and the change in the inflation rate in the United States.



- Source: FRED: CPIAUCSL, UNRATE

8.2 The Phillips Curve and Its Mutations (6 of 9)

- The line that best fits the scatter of points in Figure 8.4 is:

$$\pi_t - \pi_{t-1} = 7.4\% - 1.2u_t \quad (8.7)$$

which became known as the **accelerationist Phillips curve**. *expectation-augmental P.C.*

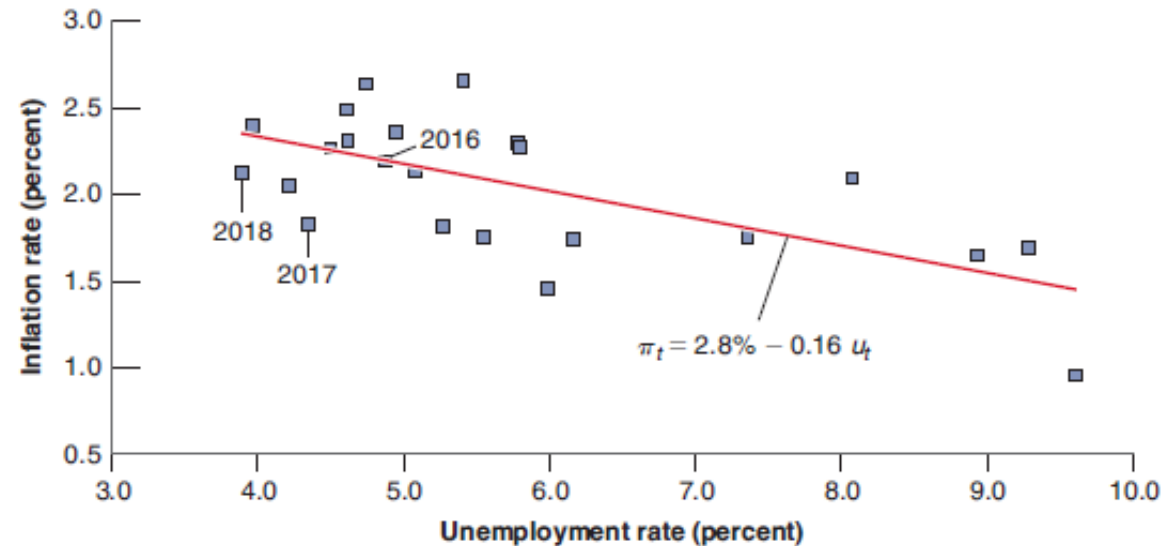
The Re-anchoring of Expectations (7 of 9)

- By the mid-1990s the Fed had largely achieved its goal of keeping inflation around 2%.
- Stable inflation changed the way people formed expectations yet again.
- Expectations of inflation that became de-anchored during the 1970s and 1980s became re-anchored in the mid-1990s.

8.2 The Phillips Curve and Its Mutations (8 of 9)

Figure 8.5 Inflation versus Unemployment in the United States, 1996–2018

Since the mid-1990s, the Phillips curve has taken the form of a relation between the inflation rate and the unemployment rate.



- Source: FRED: CPIAUCSL, UNRATE

8.2 The Phillips Curve and Its Mutations (9 of 9)

- The figure also plots the line that best fits the scatter of points for the period 1996–2018 and is given by:

$$\pi_t = 2.8\% - 0.16u_t \quad (8.8)$$

8.3 The Phillips Curve and the Natural Rate of Unemployment (1 of 2)

- Suppose $\pi = \pi^e$ and solve for u_n in equation (8.3):

$$u_n = \frac{m + z}{\alpha} \quad (8.9)$$

- Rewrite equation (8.3) as

$$\pi_t - \pi_t^e = -\alpha \left(u_t - \frac{m + z}{\alpha} \right)$$

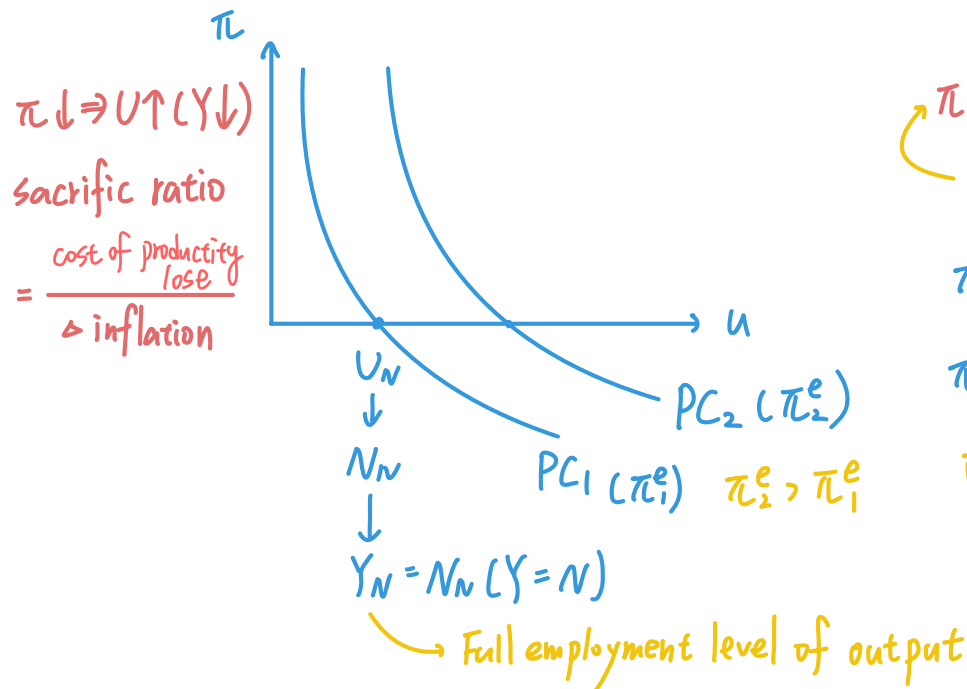
so equation (8.9) can be rewritten as

$$\pi_t - \pi_t^e = -\alpha(u_t - u_n) \quad (8.10)$$

8.3 The Phillips Curve and the Natural Rate of Unemployment (2 of 2)

Using equation (8.8) and putting $pt = 2\%$ gives:

$$2\% = 2.8\% - 0.16u_n \Rightarrow u_n = \frac{0.8}{0.16} = 5.0\%$$



$$\pi - \pi^e = -\alpha(u - u_N)$$

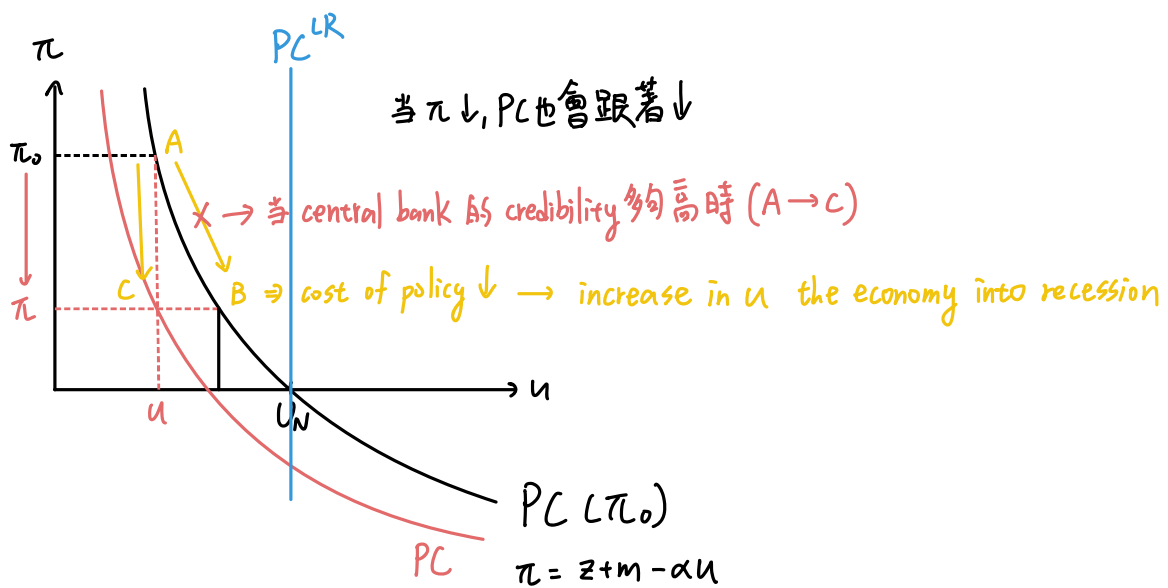
$$\pi = \pi^e + (z+m) - \alpha u$$

$$\pi^e = \pi \Rightarrow u_N = \frac{m+z}{\alpha} \text{ L.R.}$$

$$\pi^e \neq \pi \Rightarrow \text{S.R.}$$

$$\text{if } \pi^e > \pi \Rightarrow u > u_N \Rightarrow Y < Y_N$$

$$< \quad < \quad >$$



central bank AS credibility 夠高時 ($A \rightarrow C$)

central bank needs to do:

2. maximizing the employment

⇒ change the policy from time to time

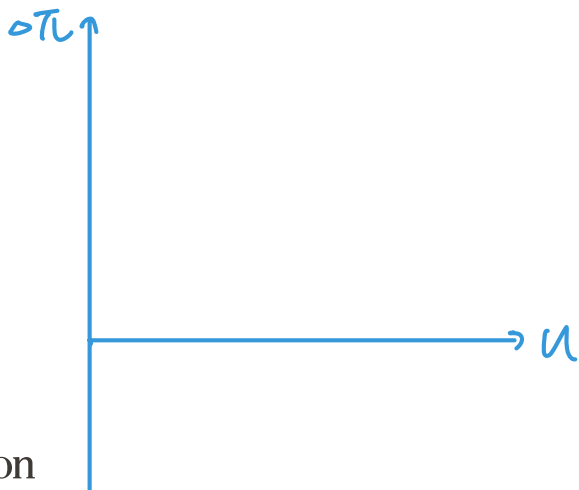
$$p \uparrow \Rightarrow u \downarrow \rightarrow y \uparrow$$

FOCUS: Theory ahead of Facts: Milton Friedman and Edmund Phelps

- Milton Friedman and Edmund Phelps argued that the trade-off between inflation and unemployment in the late 1960s was an illusion.
- Accordingly, the Phillips curve is a temporary, rather than a permanent, trade-off between inflation and unemployment that comes not from inflation per se, but from a rising rate of inflation, which results in unanticipated inflation.

8.4 A Summary and Many Warnings (1 of 5)

- The relation between unemployment and inflation in the United States today is well captured by a relation between the change in the inflation rate and the deviation of the unemployment rate from the natural rate of unemployment.
- When the unemployment rate is above (below) the natural rate of unemployment, the inflation rate typically decreases (increases).
- The natural rate of unemployment differ across countries due to, e.g., **labor-market rigidities**, and also over time.



FOCUS: Changes in the U.S. Natural Rate of Unemployment since 1990

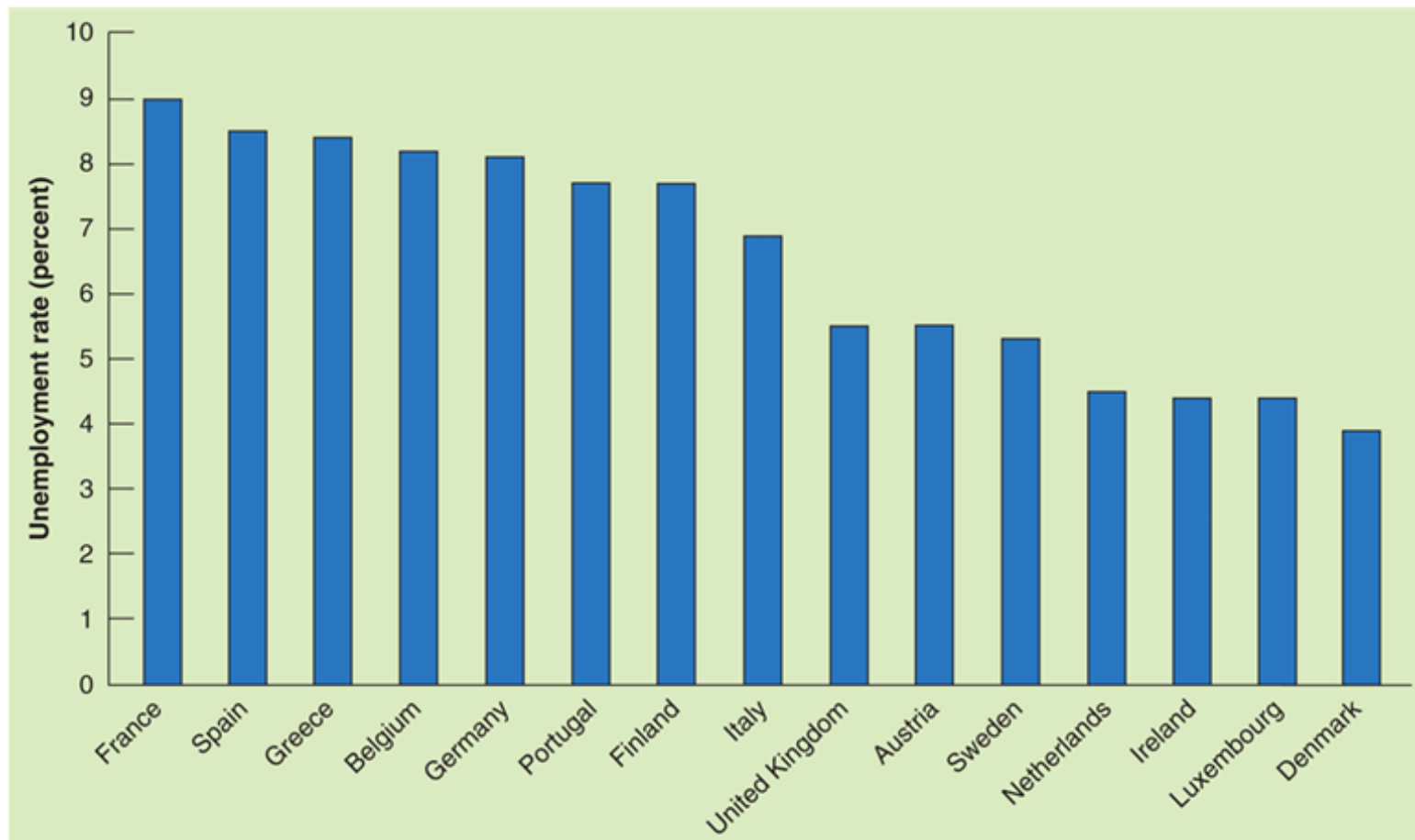
- Possible explanations for the decrease of the U.S. natural rate of unemployment from 6% or 7% in the 1970s and 1980s to about 4% today:
 - Increased globalization and stronger competition between U.S. and foreign firms may have lowered the markup.
 - Increases in employment by temporary help agencies.
 - The aging U.S. population.
 - An increase in the incarceration rate.
 - The increase in the number of workers on disability.
 - The long-term effects of the 2008-2009 recession.

FOCUS: What Explains European Unemployment? (1 of 2)

- Factors for labor-market rigidities:
 - A generous system of unemployment insurance
 - A high degree of employment protection
 - Minimum wages 非自願性失業
 - Bargaining rules, such as extension agreements
- Key facts of unemployment in Europe:
 - Unemployment was not always high
 - Some European countries actually had low unemployment prior to the start of the current crisis

FOCUS: What Explains European Unemployment? (2 of 2)

Figure 1 Unemployment Rates in 15 European Countries, 2006



8.4 A Summary and Many Warnings (2 of 5)

- When the inflation rate becomes high, the terms of wage agreements tend to change with the level of inflation.
- **Wage indexation** is a provision that automatically increases wages in line with inflation.
- Suppose λ a proportion of labor contracts that is indexed, so nominal wages move one-for-one with changes in the actual price level, equation (8.10) becomes

$$\pi_t = [\lambda \pi_t + (1 - \lambda) \pi_{t-1}] - \alpha(u_t - u_n) \quad (8.11)$$

$$\pi_t - \lambda \pi_t = (1 - \lambda) \pi_{t-1} - \alpha(u_t - u_n)$$

$$\pi_t - \pi_{t-1} = - \frac{\alpha}{1 - \lambda} (u_t - u_n)$$

8.4 A Summary and Many Warnings (3 of 5)

- When $\lambda=0$, equation (8.11) becomes equations (8.10).
- When $\lambda>0$, equation (8.11) becomes:

$$\pi_t - \pi_{t-1} = -\frac{\alpha}{(1-\lambda)}(u_t - u_n)$$
$$\pi_t - \pi_t^e = -\alpha(u_t - u_n)$$

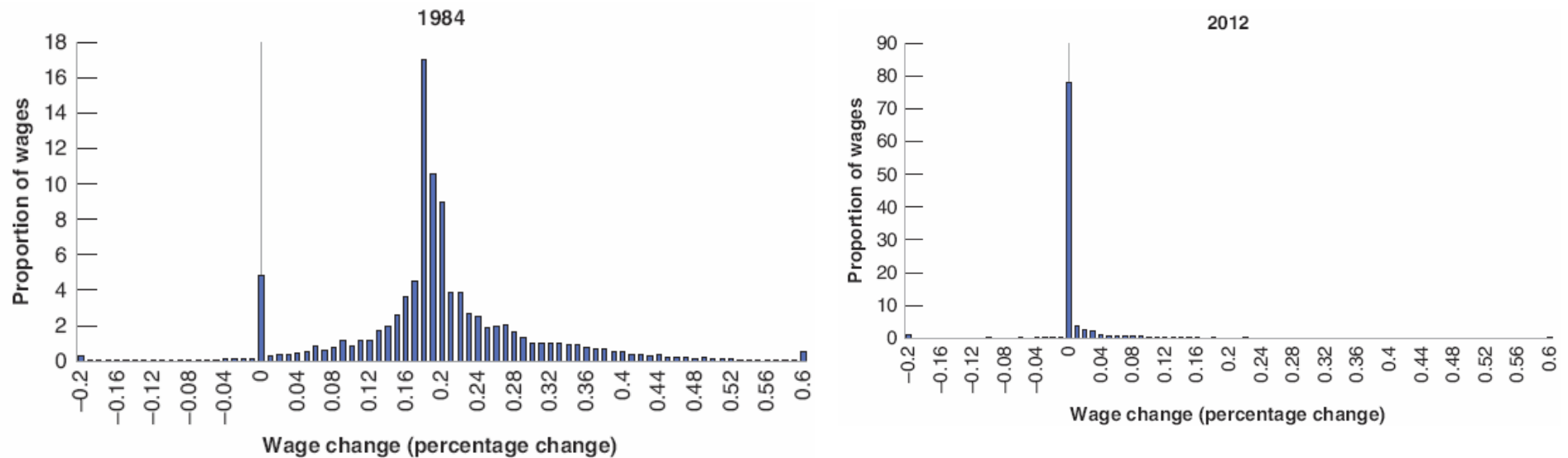
- *Wage indexation increases the effect of unemployment on inflation.*
- Without wage indexation, lower unemployment increases wages, which in turn increases prices.

8.4 A Summary and Many Warnings (4 of 5)

- When low inflation or deflation occurs, the Phillips curve relation breaks down.
- One possible reason is the reluctance of workers to accept cuts in nominal wages.

8.4 A Summary and Many Warnings (5 of 5)

Figure 8.6 Distribution of wage changes in Portugal, in times of high and low inflation



- In 1984, the inflation rate was 27%, and the distribution of wage wages was roughly symmetric.
- In 2012, the inflation rate was just 2.1%, and the distribution of wages was bunched at zero with nearly no negative wage changes.

Source: Pedro Portugal, based on Portuguese household survey.

APPENDIX: Derivation of the Relation Between Inflation, Expected Inflation, and Unemployment

- Equation (8.1) with time subscripts becomes:

$$P_t = P_t^e(1 + m)(1 - \alpha u_t + z)$$

- Divide both sides by P_{t-1} :

$$\frac{p_t}{p_{t-1}} = \frac{p_t^e}{p_{t-1}^e}(1 + m)(1 - \alpha u_t + z) \quad (8A.1)$$

- Rewrite P_t/P_{t-1} as $1 + \pi_t$ and do the same for the expected inflation rate, equation (8A.1) becomes:

$$(1 + \pi_t) = (1 + \pi_t^e)(1 + m)(1 - \alpha u_t + z)$$

$$\frac{(1 + \pi_t)}{(1 + \pi_t^e)(1 + m)} = 1 - \alpha u_t + z$$

- If the left side is not large, it becomes $1 + \pi_t - \pi_t^e - m$ and so:

$$\pi_t = \pi_t^e + (m + z) - \alpha u_t$$